DATAWORKS CHALLENGE 2026 · TRACK 3

History Matching & the 2008–2028 Forecast

What T3 did, what we found, and what is still uncertain

Reference deck for the team PPT · All T3 numbers are proxy-estimated (not simulator runs)

How T3 works: four steps

STEP 1

Search

Test 50,000 parameter sets with the proxy model

STEP 2

Select

Keep 10 sets that fit history well and differ from each other

STEP 3

Forecast

Extend oil, gas and water to 2028 from the real 2008 start

STEP 4

Range

P10/P50/P90 and scenarios; fill the 10-sheet template

The proxy (from T2) lets us test far more combinations than the 100 simulated cases we were given.

Searching 50,000 parameter sets

50,000

random parameter sets tested inside the allowed ranges (seed 42)

0.0003

tie band: sets within this of the best score count as equally good

10

sets kept: the ones farthest apart from each other

“Farthest apart” uses permeability, aquifer size and fault transmissibility. Porosity stayed almost fixed (about 1.35–1.38) in all good sets.

Our 10 sets vs the organisers' best cases

Score (lower = better)	Our 10 sets (proxy)	Organisers' cases (simulator)	Reading
Best one	0.003454	Case 6: 0.003559	A tie (gap 0.000105, inside 0.0003)
Average of top 10	0.003655	0.004528	About 19% lower

Honest message: the best set is not a clear win, but as a group our 10 sets fit the history better.

Our scores are proxy-estimated; the organisers' are simulator scores, so the two are not fully comparable.

The late-history bias: total right, split wrong

Last history quarter (Oct–Dec 2007, stored at 2008-01-01), average daily rate, our 10 sets vs the real field:

Oil

+12–15%

too high

Water

−10–11%

too low

Oil + water

<1.5%

off: the total is almost right

T1 saw the same direction in the simulator’s top-10 cases; the proxy learned it from the simulator. Gas behaves like oil (too high).

Can other parameters fix the bias? We checked: no

From the same 50,000 sets, we judged **only 2005–2008** (oil, gas, water) and picked the sets that fit those years best.

Measure	Our 10 sets	Best sets for 2005–2008 only
Late error (2005–2008)	about 6–7%	about 3.5–4.5% (halved)
Last quarter, oil	+12–15%	still +3.5–7%
Overall score (1998–2008)	about 0.0035	0.025–0.036 (7–10× worse)

Fitting the late years breaks the earlier years, so keeping our original 10 sets is the best choice.

Cold test: choosing the forecast methods



Forecast the hidden years, then compare. Also tried hiding 2 and 4 years.

Oil and gas

Arps decline curve with b = 1, fitted to 2003–2008 rates

Water

3 methods compared. Chosen: water-oil ratio vs cumulative oil (error about +15%; the others about +51%)

Limits: the test compares with the proxy curve, so it cannot see the proxy-vs-real gap. It covers 2–4 years; the real forecast is 20, so the test is optimistic.

Building the 2008–2028 forecast

Start point

The real 2008 cumulative, not the proxy's: oil 38.4M STB, gas 13.9M MSCF, water 12.8M STB

Oil and gas

Arps decline (b = 1) fitted to 2003–2008 rates, projected quarter by quarter

Water

Water-oil ratio trend, with a total-liquid cap of 13,248 bbl/day; above it, water = cap − oil

Why a cap? Without it, oil + water keeps reaching new highs (above recent levels from about 2012). Without new equipment that is not realistic. The cap is our own assumption.

Three scenarios, one in the template

IN THE TEMPLATE

Main

The method on the previous slide, with the water cap

COMPARISON

Rate-calibrated

Each stream scaled by its 2008 real ÷ proxy ratio: oil and gas × ~0.87, water × ~1.11

COMPARISON

Uncapped

Water with no total-liquid cap

The two comparisons are not “better answers”: they show how much the forecast moves if an assumption changes.

P10 / P50 / P90: the convention and a warning

At each date, take the 10 forecasts:

P10	HIGH (optimistic)
P50	middle
P90	LOW (conservative)

WARNING

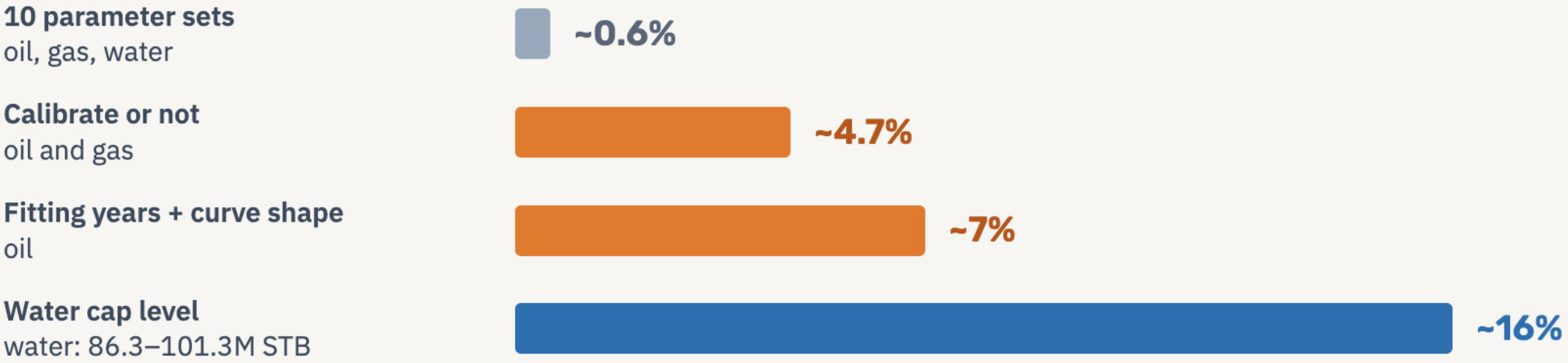
~0.6%

is all the 10 sets differ by at 2028. All 10 match history closely, and the decline curve only sees that history, so they cannot disagree.

Ten people with the same ruler agree; that does not prove the ruler is right.

Where the real uncertainty is

How much the 2028 cumulative moves when one choice changes:



Assumptions matter far more than parameters. scenario_low / high cover calibration and the cap, not the fitting years, so even that range is still too narrow.

Quality checks

- **Template filled:** 10 sheets, one row per quarter, 2008–2028 (main scenario)
- **Clean re-run:** all 8 T3 scripts from a fresh folder give exactly the same results
- **Tests:** the repository's 36 existing tests (by T1 and T2) still pass
- **Data rules:** only Case 6 is loaded for comparison; test cases 86–100 are never used

Code is in PR #3 with a cross-track review. The filled template is kept out of Git (confidential data) and shared with the team separately.

Limitations and next steps

What remains

- Oil may be too high, water too low (the late bias)
- The water cap is our own assumption
- The cold test covers 2–4 years, not 20
- Gas fitting-year sensitivity not checked

Next steps

- Run the 10 sets in the physical simulator to 2028: only this shows real parameter differences
- Add the fitting-year effect to the scenario range
- Check gas the same way as oil

No real data exist after 2008, so how far the forecast is off cannot be measured yet.

Three messages for the story

1. Our 10 sets fit the history about 19% better on average than the organisers' top 10 (proxy-estimated).
2. The late bias (oil high, water low) comes from the simulated cases; we checked that no parameter set can fix it.
3. The real uncertainty comes from our assumptions, not the parameters, so we report scenarios as well as P10/P50/P90.